

Dynamic RU-DU Reconfiguration in 6G Access Networks Using Fast Controlled Photonic Switches with 400G ZR+ Optics

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Abstract We demonstrate a novel 6G access network for dynamic RU-DU reconfiguration combining fast-controlled SOA-based photonics with 400G ZR+ optics. Cross-layer coordination achieves 206 μ s end-to-end reconfiguration latency, with scalable performance across multiple nodes and < 3.8 dB penalty at pre-FEC 10^{-2} BER. ©2026 Assad Bahari, Ali Mefleh, Nicola Calabretta

Introduction

Next-generation 6G access networks are increasingly conceived as distributed and cloudified radio access networks (RANs), where fronthaul must interconnect geographically distributed radio units (RUs) and distributed units (DUs) while meeting high capacity and stringent latency requirements of emerging services such as hyper-reliable and low-latency communications (HRLLC)[1]. Coherent optics enable high-capacity transmission over extended distances[2]. Fronthaul networks must also support dynamic operation to address failures, service degradation, load balancing, and latency optimization, as well as 6G paradigms such as cell-free RAN, where clusters of RUs cooperatively serve user equipment without fixed cell boundaries[3].

To realize such operations, fronthaul requires not only high-capacity, but also dynamic reconfiguration with low latency. Fast (sub-microsecond) optical switching is a key enabling data-plane technology to support low-latency reconfiguration directly at the physical layer[4]. Such fast reconfiguration also requires a fast control plane for state updates and consistent coordination between optical, L2, and L3 layers across nodes.

Several techniques have been proposed to accelerate the control plane, such as P4

offloading forwarding decisions to the switch ASIC[5], edge-distributed SDN controllers shorten control loops[6], and FPGA-based supervisory control to bypass software stacks for physical-layer reconfiguration [7]. Each of these approaches reduces latency only at L2 and L3 packet-layer or at optical layer. Recently an integrated optical and packet layers control was demonstrated [8], but it still operates at time-scales unsuitable for low-latency fronthaul reconfiguration.

In this paper, we demonstrate a novel high-capacity dynamic RU-DU distribution network using fast controlled SOA-based optical switches and coherent 400G ZR+ optics. The low-latency end-to-end reconfiguration exploits a fast cross-layer controller that integrates control messaging over a dedicated optical supervisory channel (OSC) to jointly coordinating optical path reconfiguration, L2 MAC forwarding, and L3 IP addressing updates. Results show successful reconfiguration across two RU-DU path with a reconfiguration latency of 206 μ s and < 3.8 dB optical power penalty at pre-FEC 10^{-2} BER.

Network Architecture and Control Plane

The proposed architecture, shown in Fig. 1, consists of distributed RUs, sparsely deployed DUs,

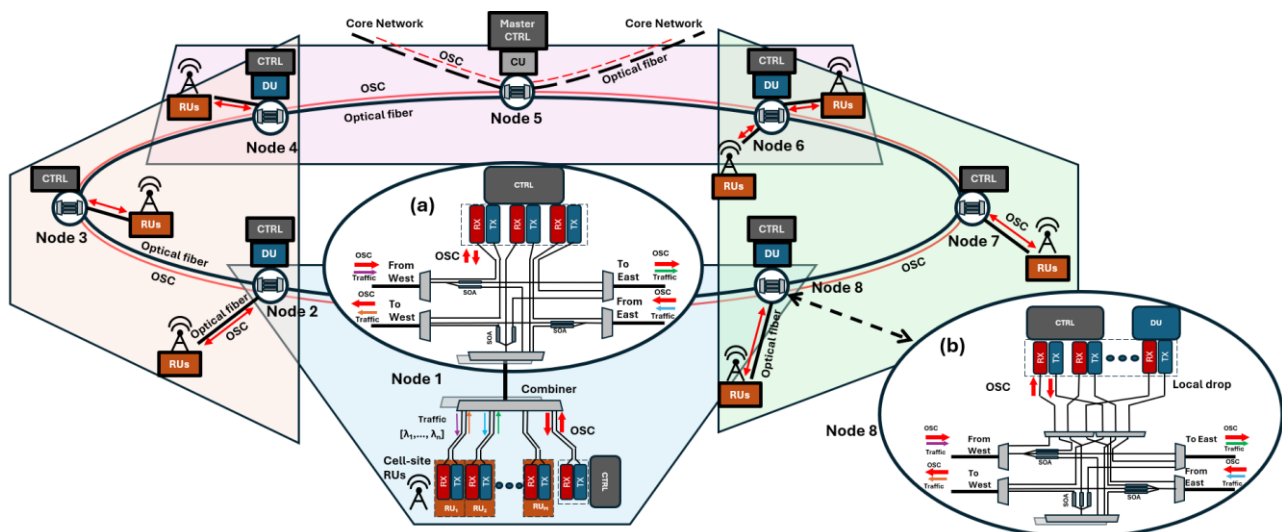


Figure 1: Proposed access network architecture depicting fronthaul sectors with dynamic RU-DU mapping.

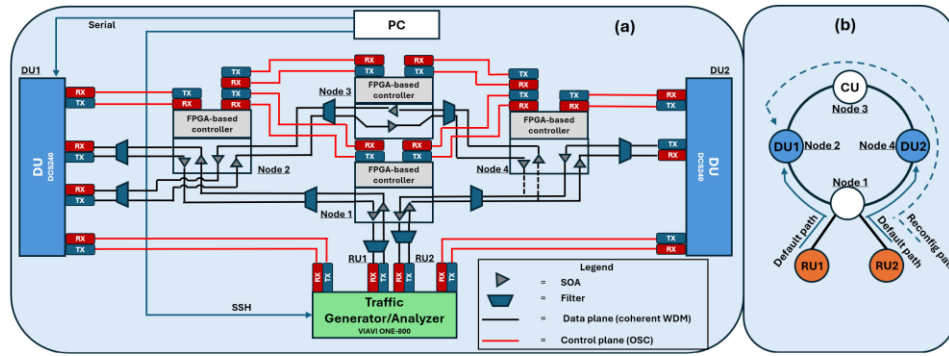


Figure 2a: Physical experimental setup. **Figure 2b:** RAN topology and depiction of default and reconfiguration paths.

and a centralized unit (CU), interconnected by SOA-based photonic switches. The network is divided into sectors (distinct colours in Fig. 1). Each sector contains at least 2 DUs for redundancy/protection and the RUs. Because fronthaul traffic remains within its sector, wavelengths from RUs transceivers can be reused across sectors, relaxing the wavelength budget. Sparse DU deployment enables centralized processing, resource pooling, and redundancy. RUs can be directly connected to the DU node or transparently across photonic nodes. RU traffic is transported transparently over C-band tuneable coherent transceivers through the photonic nodes towards the DUs, supporting multiple RU-DU associations within bespoke fronthaul sectors constrained by wavelength availability and latency. The RU-DU mapping is determined by the CU, consistent with the 3GPP CU-DU control hierarchy[2]. Each photonic node implements a multi-degree switching architecture based on AWGs and SOAs, operating as a reconfigurable OADM with per-wavelength add, drop, and directional routing across east-west links and access branches (Fig. 1a and Fig. 1b). Moreover, the SOAs provide fast (nanosecond) switching and loss compensation.

Cross-layer Control Integration

A master controller located with the CU maintains a global view of the network, including topology, wavelength allocation, switching, RU-DU mapping, and L2/L3 addressing. The control plane is hierarchical, with the master controller performing coordination and decision-making, while local node controllers handle time-critical switch configuration and control message processing. Coordination is enabled via a bidirectional OSC which operates as an out-of-band control plane extending from the CU to the RU endpoints, allowing fast and reliable exchange of control information independent of the data plane.

The OSC carries structured control messages containing node identifiers, port and link status, wavelength usage, and L2/L3 addressing information. These messages are used to signal reconfiguration events, distribute updated RU-DU

associations between the local node controller and master controller, and coordinate the establishment of new optical paths followed by corresponding updates to MAC forwarding and IP addressing to maintain end-to-end connectivity between the affected RU-DU pairs. Moreover, the OSC is also utilized as a feedback mechanism that enables a closed-loop control-plane between the master controller and local node controllers. During the initialization phase, nodes advertise their local state and receive configuration parameters from the master controller, while exchanging link-level state information with adjacent nodes over the OSC. This includes retrieving coherent transceiver capabilities via EEPROM interfaces over CMIS, such as supported wavelength ranges, grid spacing, and supported transmit power ranges, as well as receiving configuration parameters from the master controller, including assigned wavelengths (frequency and grid), transmit power setpoints, and matching endpoint parameters required to align both ends of the coherent optical link. After initialization, it enters state acquisition where it monitors optical telemetry and advertises this over the OSC towards adjacent nodes and the master controller.

Optical paths reconfiguration and RU-DU associations can be triggered either locally or by the master controller. Local triggers include events such as link failures or loss-of-signal conditions, while master-triggered reconfiguration is driven by network-level objectives such as DU reassignment, load balancing, and latency optimization. Local trigger reconfiguration actions are propagated via the OSC to adjacent nodes and the master controller, ensuring consistent network state, preventing wavelength contention at the optical layer, and maintaining end-to-end connectivity through coordinated updates of optical switching state, MAC forwarding entries, and IP addressing for the affected RU-DU traffic flows.

Experimental Setup and Results

The experimental setup is shown in Fig. 2a, with the corresponding RAN topology in Fig. 2b. The

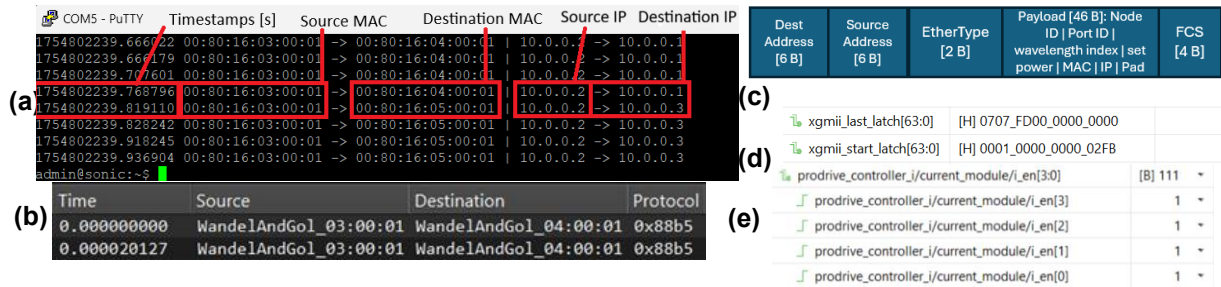


Fig. 3: a) PuTTY output showing real-time MAC/IP updates with timestamps. b) Wireshark output indicates configuration latency for OSC processing in the DCS240 at 20.127 μ s. c) Ethernet frame with payload details and field sizes (64 B frames) d) Vivado UI displaying the sent latched OSC Ethernet frame (xgmii_last_latch) e) Vivado UI displaying SOA control status parameters (i_{en} , 1=on, 0=off)

system consists of four photonic nodes, each comprising an SOA-based optical switch controlled by an FPGA (HTG-830, Xilinx UltraScale), which includes pluggable 10G SFP transceivers at 1550.12 nm (~ 0 dBm) for the OSC. The SOAs operate in linear regime with bias currents between 40–100 mA that compensate the node loss. Each RU traffic is emulated using a VIAVI OneLab Pro platform with tuneable 400G ZR+ coherent pluggables at 1561.42 nm (RU1) and 1560.61 nm (RU2) with -7 dBm launch power. As the VIAVI platform lacks native SFP interfaces, the OSC signal is carried over a separate coherent link. Each RU has a default and reconfiguration path to the DUs of the sector: RU1 has a default path to DU1 and RU2 to DU2, and can be dynamically reconfigured (RU1 to DU2, RU2 to DU1). An Edgecore DCS240 serves as DU endpoint connected to a PC to read logs and dump-files in order to verify output by observing the changed MAC and IP destinations.

The cross-layer control is validated by configuring coherent pluggables via the OSC and establishing end-to-end connectivity through L2 MAC and L3 address distribution. Control information is transmitted over the XGMII interface using a dedicated EtherType and custom payload (Fig. 3c and Fig. 3d), including node ID, port, wavelength, optical power settings, and associated L2/L3 information, which is used to configure the forwarding state and enable end-to-end connectivity.

Secondly, the end-to-end reconfiguration

latency is evaluated. The CU triggers the RU-DU reassignment via the OSC, after which the new MAC and IP addressing are distributed to the RU and DU host interfaces (red boxes in Fig. 3a) and the SOA states are switched to establish the new optical path (Fig. 3e). The measured end-to-end reconfiguration latency, identified from the inter-flow transition in the timestamped packet dump is approximately 206 μ s, including the OSC processing time of 20 μ s in the DU (DCS240), and 20 μ s in the RU (VIAVI) to assign the new MAC/IP mappings (see Fig. 3b)

Lastly, scalability and optical performance is assessed by transmitting traffic through multiple photonic nodes under both default and reconfigured paths. In the first scenario, the paths RU1-Node1-Node2-DU1 and RU2-Node1-Node3-DU2 are evaluated. The resulting pre-FEC BER is shown in Fig. 4a, where both paths exhibit comparable performance. In the second scenario, the photonic switches are reconfigured to establish the new optical path as seen in Fig. 2b. The resulting optical performance is shown in Fig. 4b, where a progressive power penalty is observed with increasing node count, reaching approximately 0.7 dB after one node and approximately 3.8 dB after four nodes. This corresponds to a decrease in OSNR to 31.2 dB after four nodes.

Conclusion

We demonstrated dynamic RU-DU reconfiguration in a disaggregated 6G access network combining SOA-based photonic switching with 400G ZR+ coherent optics and cross-layer control. Reconfiguration was successfully executed across all tested RU-DU configurations, with coordinated updates spanning optical path switching, L2 MAC forwarding, and L3 IP addressing distributed via a dedicated OSC. The end-to-end reconfiguration latency was approximately 206 μ s. Pre-FEC BER on the order of 10^{-2} , with a cumulative power penalty of 3.8 dB was achieved over four photonic nodes and 31.2 dB OSNR, confirming scalable optical performance under dynamic operation.

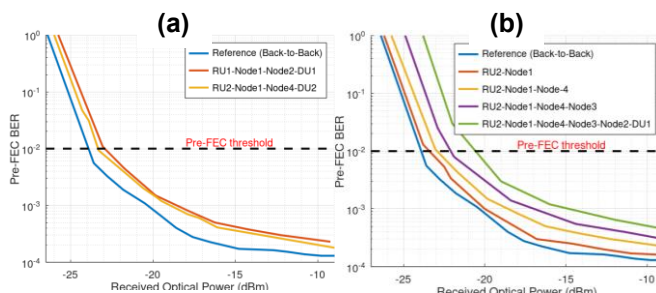


Figure 4a: Pre-FEC BER vs received optical power link comparison **b):** Pre-FEC BER vs received optical power re-configuration link

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